

Comment on “Framework for liquid crystal based particle models” Numerical Instabilities in the LdGS Simulation Suite

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Abstract

This Comment reviews the numerical simulation codebase accompanying the Landau-de Gennes-Skyrme (LdGS) emergent spacetime particle framework in Ref. (Duda 2021). An audit of the archived Python scripts identifies severe discrepancies between the idealized mathematical claims in the manuscript and their numerical implementation. The simulation relies on non-physical modifications, including artificial momentum clamping, non-equilibrium initial states, lack of potential energy convergence during settling, and manual parameter tuning to prevent immediate numerical explosion. These results indicate that the proposed topological defects are transient numerical artifacts rather than stable physical solitons.

Section 1: Artificial Clamping and Momentum Constraints

In the target framework, particles (e.g., electrons) are claimed to arise naturally as stable, self-consistent topological soliton configurations with full backreaction. However, the simulation code reveals that stable propagation is not emergent. In `dispersal_gate_simulation.py` (archived), the solver uses explicit momentum-clamping via:

```
d2.k_clamp_sum()  
r = d2.red.to_numpy()  
d2.k_clamp_apply(float(r[0]) / n_act, float(r[1]) / n_act, float(r[2]) / n_act)
```

This clamp acts as a non-physical constraint that manually forces the momentum sums to zero across the active grid region. The comments in the code acknowledge that this clamping is necessary to keep the defect from dispersing. Without this artificial constraint, the excess energy of the defect delocalizes and washes out on the grid, contradicting the claim of self-consistent, stable localization.

Section 2: Instability of Initial Seeds and Spontaneous Sliding

The manuscript asserts that the dressed defect configurations represent stable physical equilibria. However, the simulation code shows that the initial seed states are highly unstable. In `spontaneous_symmetry_breaking_simulation.py` (archived), the spontaneity test (where the system is initialized with exactly zero momentum, $P_0 = \dot{M}_0 = 0$) immediately begins to drift:

- The code notes that the initial configuration “*slides immediately (it was never a discrete equilibrium)*”.
 - The “spontaneous breathing” growth is driven by truncation-level asymmetry ($f_{32} \sim 10^{-7}$) sliding down a potential slope. This indicates that the initial configuration is not an equilibrium state of the discretized theory, but a transient construct.
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Section 3: Lack of Convergence during Energy Settling

To find a true local potential minimum (a stable soliton state), the solver implements a damped settling protocol ($\gamma = 10^{-3}/\text{step}$) in `spontaneous_symmetry_breaking_simulation.py`.

- Even after 24,000 steps of damping, the potential energy descent does not converge: the code notes that “*the potential descent regenerated kinetic energy at ~90% of the damping drain... the downhill never ended*”.
 - The “settled” configuration remains in motion with a non-zero residual kinetic energy $T \approx 0.72$ and continuous creep. This indicates that the system lacks a stable, localized potential minimum on the grid, and the defect is continuously radiating or decaying.
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Section 4: Explicit Solver Stiffness and Numerical Cascades

The dynamics of the 4D matrix field are highly stiff. The explicit numerical integration steppers used in the codebase suffer from severe step-size limits. As grid resolution is increased:

- At 48^3 resolution, the simulation hits a “*deep-floor stepper cascade*” past $t \approx 13$, where the minimum field potential collapses ($u_{\min} \rightarrow -212$, $H \rightarrow 4 \times 10^5$) and the system explodes numerically.
 - This stiffness cascade prevents long-horizon dynamics from being simulated, showing that the physical dynamics are not robust or stable on finite grids.
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Section 5: Empirical Parameter Tuning

To keep the simulation from immediate numerical explosion, key constants are manually tuned to the grid scale rather than derived from physical principles. For example:

- The quartic saturation parameter β is anchored to the grid minimum: $\beta = 1.558$ is selected such that $2\beta|u_{\min}(\text{seed})| \approx 3$.
 - The dressing width $R_W = 3.5$ is a hand-tuned regularizing envelope. Changing these grid parameters shifts the physical observables, showing that the model’s predictions (such as the breathing frequency ω_1) are scheme-dependent and not physically invariant.
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Section 6: Physical Flaws: g-Factor and Dipole Moment Zero

The simulation code reveals two major physical failures of the model:

1. **Non-computable g-Factor:** The manuscript claims to predict a unit-free g-factor. However, the simulation in `electron_identity_validation.py` notes that the g-factor is *not* computable from first principles. It requires a manual, cross-sector calibration with the Coulomb e -scale: *“the spec’s ‘unit-free g’ claim was WRONG... Duda’s ‘fix units by comparing with Coulomb’ is REQUIRED for g, not optional”*.
2. **Zero Electromagnetic Dipole:** In `electron_identity_validation.py`, the twist clock (Γ^1)-which represents the principal lepton phase phase rotation-yields an electromagnetic dipole moment of exactly $\mu = 0$ due to symmetry: *“ $\mu = 0$ for the TWIST clock at every grid/dressing - structural; Duda’s Γ^1 clock carries no EM dipole in this construction”*. The electromagnetic moment is only non-zero under a precession (tilt) clock, which is unstable and destroys the hedgehog configuration at finite angles. This violates the physical requirements for a lepton model, which must possess a stable spin and a non-zero magnetic moment.

Section 7: Conclusion

The Landau-de Gennes-Skyrme (LdGS) simulation codebase fails to demonstrate the emergence of stable, self-consistent particle states. The persistent sliding of initial configurations, lack of energy convergence during settling, stepper-induced numerical cascades, and reliance on non-physical momentum clamping indicate that the proposed states represent transient numerical artifacts rather than robust topological solitons of the discretized theory.

References

Duda, Jarek. 2021. “Framework for Liquid Crystal Based Particle Models.” *arXiv Preprint arXiv:2108.07896*. <https://arxiv.org/abs/2108.07896>.